Reusing the CRT index in a Diophantine consistency equation

An explicit quartic in 215 unrestricted integer variables

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Abstract

We construct an explicitly specified polynomial $P \in \mathbb{Z}[x_0, \ldots, x_{214}]$ of degree four such that ZFC proves that P has no integer zero if and only if ZFC is syntactically consistent. The polynomial is the sum of 154 squares of quadratic polynomials, printed in the appendix and supplied as exact coefficient data. Its collected expansion has 5660 nonzero monomials and coefficient height 91679.

The new arithmetic step eliminates a derived-formula-size test. The independent CRT row-index representative already has a Pell pair and can be chosen even and arbitrarily large in its residue class. It therefore also supplies the semantic index bound used by the shared Pell certificate. The removed size test is compensated by a larger label-weight parameter, certified with a polynomially constructed exponent. A sharp elementary product-separation lemma proves that this change preserves proof decoding. Five exact quadratic projections and one additional affine projection reduce the arithmetic source from 221 to 215 coordinates. Reversible coordinate and row changes then reduce its coefficient height as well as its number of monomials. The resulting 215-variable quartic is the principal equation. The construction uses the numerical C_{222} calculus directly. The inherited logical calibration is identified explicitly; no new end-to-end Lean or Rocq formalization is claimed.

1 Result, comparison, and precise scope

The principal equation is the following single ordinary Diophantine equation:

$$\boxed{P(\boldsymbol{x}) = \sum_{j=0}^{153} R_j(\boldsymbol{x})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{215}.} \tag{1}$$

Every R_j is an explicitly given quadratic polynomial with integer coefficients. Appendix A prints the entire list. The file `equations/principal/equation.json` defines the same list without abbreviations, and its neighboring expansion contains every collected coefficient of P. These files, the variable order, and the integer domain are part of the definition. There is no uninstantiated universal polynomial, machine index, existence-defined coefficient, or variable exponent in P.

Theorem 1.1. *Using the identified inherited arithmetic and logical calibration, the construction and proofs below establish*

$$\mathrm{ZFC} \vdash \Big(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{215}\ P(\boldsymbol{x}) = 0\Big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Here consistency is the absence of a finite proof of falsity in the effective ZFC presentation used by the repository's calibration.

The theorem is an equivalence, proved without assuming either consistency or inconsistency. It is not a proof that the displayed polynomial has no zeros. The traditional mathematical proof is separate from the finite artifact checks. The latter authenticate coefficients, polynomial identities, and reconstruction maps; they do not prove unbounded Pell or logical calibration theorems by testing examples.

Form	Variables	Squares	Degree	Monomials	Height
Pinned parent	225	162	4	7,867	128,446
Principal	215	154	4	5,660	91,679
Companion	214	153	4	5,486	1.0996×10^{12}
Companion	213	152	4	5,402	2.4179×10^{24}

The companion heights are rounded upward here. Their exact values are

$$1\,099\,511\,627\,852 \qquad \text{and} \qquad 2\,417\,851\,639\,242\,452\,488\,945\,685.$$

The comparison is pinned to ProveIt commit `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`. The matched parent is the 225-variable pooled-power quartic in source 158 [1]. Sources 151 and 157 supply the factorial-free binomial CRT and affine beta constructions [2, 3]; source 159 supplies the exact quadratic projection method [4]. The present result changes the choice and role of the semantic anchor and removes the derived-label range, rather than merely reevaluating an old polynomial. It also supplies a precise endpoint obtained by composing those changes with exact algebraic reductions.

"Simplest" is not a single ordered mathematical criterion. Variable count, total degree, coefficient height, number of monomials, and circuit size can improve in different directions. The repository contains substantially lower-dimensional equations in other degree/coefficient regimes. We claim a concrete improvement in the directly compiled quartic family, not a global minimum or worldwide priority for the general idea.

2 The retained proof-calculus architecture

The numerical calculus has truth code 0, falsity code 1, modus ponens, generalization, eight logical axiom schemata, and a guarded seed schema. Its literal square-shell constructors are

$$I(a,b) = 2((a+b)^2 + a), \qquad M(a,b) = I(a,b) + 1,$$
$$A(a,b) = 2((a+b)^2 + 3a + 4b + 3), \qquad E(a,b) = A(a,b) + 1.$$

They encode implication, membership, universal quantification, and equality. Put $N(a) = I(a,1)$ and $X(a,b) = N(A(a,N(b)))$. The pairing maps $Q_0 = (a+b)^2 + a$ and $Q_1 = (a+b)^2 + 3a + 4b + 3$ enumerate complementary square shells; parity supplies the four constructors. The inherited decoding theorem transports finite refutations between these codes and the original Cantor-coded C_{222} calculus, fixing 0 and 1 [7].

For clarity the retained logical codes are

$$\begin{aligned}
r_1 &= X(a, E(a,b)),\\
r_2 &= I(A(a, I(b,c)), I(A(a,b), A(a,c))),\\
r_3 &= I(E(a,b), I(M(c,a), M(c,b))),\\
r_4 &= I(A(a, A(b,c)), A(b, A(a,c))),\\
r_5 &= I(E(c,a), I(M(c,b), M(a,b))),\\
r_6 &= I(X(a, A(a,b)), b),\\
r_G &= I(E(c,a), I(E(c,b), A(M(a,b), E(a,b)))),\\
r_8 &= I(I(I(c,d), a), I(I(a,c), I(b,c))).
\end{aligned}$$

Define

$$\mathrm{Req}(b,k,m) = A(b, I(m, N(A(b, I(m, N(M(b,k))))))),$$

$$\begin{aligned} S(a,b,k) &= A(k, I(M(k,a), M(k,b))), \\ T(a,b,k,m) &= N(A(a, I(S(a,b,k), \mathrm{Req}(b,k,m)))), \\ \mathrm{Clos}(a,b,k,m) &= A(b, I(M(b,k), N(A(a, I(m, T(a,b,k,m)))))), \\ \mathrm{nks}(a,b,c) &= A(c, I(\mathrm{Clos}(a,b,c,c), N(\mathrm{Req}(b,c,c)))), \\ \mathrm{RV}(a,b,c) &= I(M(a,b), A(c, M(a,b))), \\ r_9 &= N(I(\mathrm{RV}(a,b,c), \mathrm{nks}(a,b,c))). \end{aligned}$$

The last schema is allowed when $c \neq a, b$. Its arithmetic guard is

$$G = (c-a)^2(c-b)^2 - (d+1)^2 = 0, \qquad a, b, c, d \in \mathbb{N}.$$

Such a d exists exactly when the two disequalities hold. The parameter d can separately be an ordinary formula parameter in the r_8 branch. The alternatives do not require both interpretations simultaneously.

The proof transcript is encoded by positive weights $w(j) = 1 + Fj$. A backward axiom step has $S'w(r) = S$, modus ponens has $S'w(b) = Sw(a)w(I(a,b))$, and generalization has $S'w(A(a,b)) = Sw(b)$. Truth uses $S' = S$. Let $D_r = S'w(r) - S$ and define the analogous MP and generalization differences. The retained nonnegative local polynomial is

$$\mathcal{L} = (S'-S)^2 \prod_{r \in \mathcal{R}} D_r^2 (D_{r_9}^2 + G^2) D_{\mathrm{MP}}^2 D_{\mathrm{gen}}^2 + B_0^2 + B_1^2.$$

Here $\mathcal{R} = \{r_1, r_2, r_3, r_4, r_5, r_6, r_G, r_8\}$, and B_0, B_1 are the two beta-remainder equations. Thus $\mathcal{L} = 0$ means that one valid proof-rule branch and both beta equations hold. The principal construction retains this local polynomial unchanged.

3 Formula bounds without a range test

The old arithmetic source separately required a derived sum of formula labels to lie below the proof horizon in every local row. That required its own numerator power, binomial divisibility test, and Pell index. The replacement needs only a fixed, much more generous syntactic bound.

Lemma 3.1 (Constructor-depth bound)**.** *For $t \geq 16$, every formula label needed by a local rule with $0 \leq a, b, c, d < t$ is smaller than $t^{2^{20}-1}$. In particular it is at most $L = t^{2^{20}}$.*

Proof. For a positive integer s, if integer inputs $0 \leq x, y < s$, then $x, y \leq s-1$ and each of $I(x,y), M(x,y), A(x,y), E(x,y)$ is less than $8s^2$. For the largest constructor, $A(s-1, s-1) = 8s^2 - 2s$, and E adds one. If the inputs are less than t^e, $e \geq 1$, each output is consequently less than t^{2e+1}. Starting with exponent one, a term of constructor depth h is bounded by $t^{2^{h+1}-1}$.

The exact syntax trees of the nine axiom codes have depths

$$4, 3, 3, 3, 3, 5, 4, 3, 19.$$

The MP and generalization labels have depth one; parameter labels have depth zero. The depth certificate checks the trees against the pinned literal constructor source, including all occurrences of N and X. The maximum depth nineteen proves the result. No asymptotic estimate or coefficient rounding enters this argument. □

Lemma 3.2 (Sharp product separation)**.** *Let L, t, F be positive integers with $L \geq t \geq 1$ and $F \geq L^t$. Consider two finite lists of positive integer labels at most L. Suppose the first has at most t entries and the second at most $2t+1$ entries. If their products of weights $1 + Fj$ agree, their label multisets agree.*

Proof. First the lists have equal length. If their lengths differ, let the smaller length be $r \le t$. The product for the smaller list is at most $L^r(F+1)^r$, while the larger is at least $(F+1)^{r+1}$. But $F+1 > L^t \ge L^r$, a contradiction.

Write their common length as $m \le t$. For $1 \le k \le m$, each elementary symmetric coefficient e_k of the labels satisfies

$$1 \le e_k \le \binom{m}{k} L^k \le m^{m-k} L^k \le L^m \le F.$$

The middle estimate uses $\binom{m}{k} = \binom{m}{m-k} \le m^{m-k}$. Thus the corresponding coefficients of the difference of $\prod(1+jX)$ have absolute value strictly less than F; the constant coefficient is zero. If the polynomial difference were nonzero, divide its value at F by the lowest power with nonzero coefficient and reduce modulo F. A nonzero integer of absolute value less than F would be divisible by F, impossible. The two polynomials coincide. Their linear factors over $\mathbb{Q}$ recover every label with multiplicity. The empty-list case is immediate. Labels zero, if present, contribute unit factors and are discarded before this argument. □

The boundary $F = L^t$ is permitted. Bounding coefficient differences by twice the largest coefficient would miss this useful fact: both coefficients are positive, so their difference has absolute value at most $F-1$.

The new principal semantic choices are

$$t = 128 + 4z^2, \qquad T_2 = t^2, \qquad K = T_2^2 = t^4, \qquad F = t^K. \tag{2}$$

Since $t^3 \ge 128^3 = 2^{21} > 2^{20}$, Lemma 3.2 applies with $L = t^{2^{20}}$. Only the first three relations in (2) are written as direct polynomial equations. The last is certified by the retained Pell power mechanism. The exponent K requires two quadratic wires, rather than a large coefficient multiplying t. This is why the principal retains small coefficients.

4 The factorial-free arithmetic front

Use the following semantic quantities:

$$B = \binom{F+t}{t}, \qquad N = B^x, \tag{3}$$

$$y = F(2v+1), \qquad C = 2c_0, \tag{4}$$

$$H = 2 + 4(C + a + b + c + d + S + S' + u + v_q), \qquad q = (BH)^x. \tag{5}$$

The nine large CRT representatives are $a, b, c, d, S, S', u, v_q, C$; their local residues encode one proof row. The independent names c_0 and v in (4) are unrestricted integer witnesses, not extra local schema parameters.

The endpoints and row-index congruence are

$$x = F + 1 + (y+1)u_0, \qquad x = 1 + [1 + (t+1)y]u_t, \tag{6}$$

$$B\eta = C + F + 1. \tag{7}$$

For a local row with index $i = C \bmod p^{ex}$, the beta equations are

$$x = S + [1 + (i+1)y]u, \qquad x = S' + [1 + (i+2)y]v_q.$$

The retained ranges are $a, b, c, d, C < t$, $S, S' \le y$, and $u, v_q < x$ on local decoded residues. There is no longer a range test on a derived formula-label sum.

4.1 Why one binomial modulus covers every row

Here is the inherited selector theorem in the form used by this construction [2]. Let $\Lambda_t = \operatorname{lcm}(1, \dots, t)$. If $R \geq \Lambda_t^2$ and $B = \binom{R+t}{t}$, then:

1. for every $1 \leq i \leq t$, some full prime power $p^e \| B$ divides $R + i$ and satisfies $p^e > t$;
2. every full prime power $p^e \| B$ divides $R + i$ for at least one such row i.

Both directions matter. The first authenticates each row; the second makes the reverse CRT construction compatible at every prime of B.

For completeness, the elementary proof is short. The valuation formula shows $t\binom{t-1}{i-1} \mid \Lambda_t$. Since $R + i > \Lambda_t^2$, some prime has $w = v_p(R + i) > 2a$, where $a = \lfloor \log_p t \rfloor$. Other factors $R + j$ have the same p-valuation as $j - i$. Consequently

$$v_p(B) = w - v_p\left(t\binom{t-1}{i-1}\right) > a, \qquad v_p(B) \leq w,$$

which proves the first statement. For the second, let $w = \max_i v_p(R + i)$. In Legendre's formula for $v_p(B)$ every summand is zero or one, and all levels above w vanish. Hence $v_p(B) \leq w$; a row attaining the maximum is compatible.

The standard elementary estimate $\Lambda_t \leq 4^{t-1}$ gives $F \geq t^t \geq 16^t > \Lambda_t^2$. Therefore $R = F$ is valid. Unlike the parent, this construction has no equation requiring a variable to equal a factorial.

4.2 The range lemma and precision

For a prime p, integers $A \geq 0$ and $k, e \geq 1$,

$$p^e \mid \binom{A}{k} \quad \Longrightarrow \quad A \bmod p^e < k. \tag{8}$$

If none of the k consecutive factors $A - k + 1, \dots, A$ is divisible by p^e, the valuation formula has at most $e - 1$ nonzero levels, a contradiction. A factor divisible by p^e gives the claimed residue. The convention $\binom{A}{k} = 0$ for $A < k$ covers the boundary.

At a selector $p^e \| B$, the range for C and (7) give the exact local index $i - 1$ in $0, \dots, t - 1$. The other ranges recover the bounded local tuple modulo p^{ex}. The retained local bound is

$$0 \leq \mathcal{L} < (2 + t + x + y + F)^{2^{23}} < 2^x \leq p^{ex}. \tag{9}$$

Its first inequality is the existing literal C_{222} circuit majorant [6]; changing the size of F does not change that polynomial bound. Once signs are recovered, $x \geq 2 + (t+1)y$ and $y \geq F$ imply $2 + t + x + y + F < 2x$ and $x \geq F + 1 > 2^{32}$. If $2^n \leq x < 2^{n+1}$, $n \geq 32$, then $2^{23}(n + 2) < 2^n \leq x$, proving the second strict inequality in (9). Thus $N \mid \mathcal{L}$ forces each selected local value to be zero.

5 A freely chosen CRT index as the Pell anchor

The principal change is independent of the detailed syntax of C_{222}. An integer used as a CRT representative can be enlarged by whole multiples of its modulus without changing the finite transcript it represents. We use that freedom to make C the largest requested Pell index.

Write $X_j(A), Y_j(A)$ for the Pell polynomials, defined by

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{j+1} = AX_j + (A^2 - 1)Y_j, \quad Y_{j+1} = X_j + AY_j.$$

Signed indices have $X_{-j} = X_j$, $Y_{-j} = -Y_j$. The square frame is $W = 2 + \sum \alpha^2$, where every actual power base, index, and proposed output occurs among the α. The retained odd Pell frame proves $|A| > W^W$ and $|A - 1| > 4W^3$.

The fourteen nominal indices are

$$t,\ F + t,\ a, b, c, d,\ S,\ y + 1,\ S',\ u,\ x,\ v_q,\ C,\ K. \tag{10}$$

The former derived-label pair is reassigned to K; the existing C pair becomes the anchor. With anchor coordinates (U, Y), retain auxiliary Pell pairs (J_X, J_Y) and (V_X, V_Y) and comparison center D satisfying

$$\begin{aligned} U^2 - (A^2 - 1)Y^2 &= 1, & J_X^2 - (A^2 - 1)J_Y^2 &= 1, \\ V_X^2 - (D^2 - 1)V_Y^2 &= 1, & J_Y &= Y^2(2e + 1), \\ D &= 1 + 4J_Y(2d_0 + 1), & D &= A + J_X f_0, \\ V_X &= U + J_X g_0, & V_Y &= C + 4Y h_0. \end{aligned}$$

The squared bound is

$$Y^2 = C^2 + 1 + \sum_{i \neq \text{anchor}} Y_i^2 + r^2 + s^2 + w(w + 1). \tag{11}$$

All these variables are unrestricted integers. Multiplication wires turn the displays into quadratic residuals; the final emitted data contain the actual wires, not a higher-degree shorthand.

Proposition 5.1 (Anchor substitution)**.** *The retained signed anchor and pooled orientation proof applies with nominal index C in place of the derived formula-label index. No monotonicity or syntactic interpretation of C is required in soundness. For completeness, C can be chosen even and larger than all other indices in* (10) *while preserving every required CRT residue.*

Proof. Equation (11) gives $|Y| > |C|$, so $Y \neq 0$. Pell classification and the retained squared-divisibility argument give an absolute index $i > 0$ with $|Y| = Y_i(|A|)$ and $i = \pm C \pmod{2|Y|}$. Since $1 \leq i \leq |Y|$ and $|C| < |Y|$, one obtains $C \neq 0$ and $i = |C|$. Every other classified index is strictly smaller than $|C|$ by the same squared bound.

The pooled orientation row is, modulo its retained norm identities,

$$\sum_i \big((X_i - 1)^2 + (Y_i - k_i)^2\big) = (A - 1)Q_0.$$

Modulo $A - 1$, write $s_i \in \{-1, 1\}$ and r_i for the small abscissa and ordinate residues of each classified pair. The absolute value of r_i is its classified index, hence at most $|C| < W$. Thus $|s_i - 1| \leq 2$ and $|r_i - k_i| < 2W$. For the fourteen pairs their squared-error sum satisfies

$$0 \leq \sum_i \big((s_i - 1)^2 + (r_i - k_i)^2\big) < 4 \cdot 14(W^2 + 1) < 4W^3 < |A - 1|.$$

It must vanish, identifying every pair with the canonical signed pair at its nominal index. This includes the new K pair and the C anchor. These are exactly the hypotheses and steps of the inherited signed-anchor proof [5]; the name or origin of the nominal index never enters.

For the reverse construction, first fix t, K, F, y, x, B, N and the prescribed residue classes of the nine representatives. None of these choices depends on the eventual representative C. Since t is divisible by four and $F = t^K$, $v_2(F) = K v_2(t) > \lfloor \log_2 t \rfloor$. Therefore $v_2(F + i) = v_2(i)$ for $1 \leq i \leq t$, and B and N^2 are odd. Each prescribed class modulo N^2 has cofinally many positive even representatives, with step $2N^2$. Choose the other eight representatives first, then choose C in its class above all the other fixed

indices. This also preserves $B \mid C + F + 1$, since $B \mid N^2$. Only afterward are H, q, W, A chosen. There is consequently no cyclic size requirement.

For canonical pairs at a positive sufficiently large A, the strict index inequality makes the slack in (11) nonnegative. Since C is even, $Y_C(A)$ is even and $X_C(A)$ odd. The inherited explicit auxiliary choice at index $CY_C(A)$ makes $J_Y/Y_C(A)^2$ odd; a two-modulus CRT then constructs D and the remaining comparison coordinates. Every nonnegative slack has the ternary representation $r^2 + s^2 + w(w+1)$. Thus all anchor witnesses exist. □

6 Power certificates and the noncircular sign argument

The polynomial retains sixteen power requests: three individual requests for $F = t^K$, $N = B^x$, $q = (BH)^x$; ten at base $q+1$ for the binomial numerators; and three at base q for q^t, q^{y+1}, q^x. The square frame has thirty-five actual atoms, including duplicates where the same integer has two roles. Its coefficient identity is independently checked against every requested base, index, and output.

For a block at nonzero base n, let $Z_i = AY_i$ and let m_i be the proposed output. The inherited pooling mechanism uses just two rows:

$$K_n - 2\sum_i Y_i(m_i - X_i) - (2A - n)L_n = 0,$$
$$\sum_i ((m_i - X_i + Z_i)^2 - Y_i^2) - nK_n + L_n = 0.$$

They imply divisibility of the sum of squared power errors by $T_n = 2An - n^2 - 1$. To handle signed indices, put $b = \lfloor \sqrt{W-1} \rfloor$ and multiply by n^{2b}. Canonical Pell congruences then give

$$T_n \mid E = \sum_i (m_i n^b - n^{b+k_i})^2, \qquad 0 \le E \le 4rb^{4b} < W^W < |T_n|.$$

Here $r \le 15$, every exponent $b + k_i$ is nonnegative, and $\gcd(n, T_n) = 1$. Consequently every output is exact; at $|n| \ge 2$ the index is nonnegative, since a negative index would force an integer times a positive power of $|n|$ to equal one. The individual requests use the same canonical power argument. All displayed exponentials occur in this proof, not as operations in the final polynomial.

It remains essential to justify the bases before interpreting these requests. The order is as follows.

1. The known $t \ge 128$ and $K = t^4$ give $F = t^K > 0$, even. Since $C = 2c_0$, equation (7) has odd right side, so $B \ne 0$. Independently, $H \equiv 2 \pmod 4$ for every signed representative tuple. Thus $|BH| \ge 2$.
2. The individual BH request gives $x \ge 0$ and $q = (BH)^x$. Equation $y = F(2v+1)$ gives $|y| \ge F$. If $x = 0$, the last endpoint would make the integer $1 + (t+1)y$, whose absolute value exceeds one, divide -1. Therefore $x \ge 1$, $|q| \ge 2$.
3. The q block is now legitimate. Its $y + 1$ index is nonnegative. A negative y would be at most $-F < -1$, so $y \ge F$. The first endpoint excludes $u_0 < 0$, and the last excludes $u_t \le 0$. Thus $x \ge F + 1$ and $x \ge 2 + (t+1)y$.
4. Hence $|q + 1| \ge 2$. Its block recovers the nonnegative signs of all nine representatives and all their exact numerator powers. Now $H \ge 2$. No earlier step needed these signs.
5. The first binomial digit authenticates $B > 0$, as proved next. It follows that $q > 0$ and $N = B^x > 0$, completing the sign recovery.

6.1 Exact recovery of the digits

Let $M = q^x$. Ten normalized remainder rows share

$$M^2 = 1 + 4\sum_{i=1}^{10} \rho_i^2 + \alpha^2 + \beta^2 + \gamma(\gamma+1). \tag{12}$$

For a denominator $D_i = q^{k_i}$, the ratio equation $D_i a_i = M$ and normalized digit equation imply $r_i = \rho_i/a_i \in \mathbb{Z}$ and the usual unscaled division equation. The norm gives $|r_i| < |D_i|/2$, even before signs of the radix are known. A true low binomial remainder with absolute value below $|D_i|/32$ must equal r_i, by congruence and the strict interval width.

For the first digit, every coefficient of $(q+1)^{F+t}$ through degree t is at most $(F+t)^t \le 2^{t(F-1)}$. On the other hand, $|q| \ge 2^x \ge 2^{(t+1)F+2}$. The true low remainder and the t-th coefficient are therefore smaller than their needed denominators divided by 32. The normalized first row gives $B \equiv \binom{F+t}{t} \pmod q$. Also $|B| \le |BH| < |q|/2$, since $x \ge 3$; the positive binomial coefficient is below $|q|/32$. This proves the exact identity $B = \binom{F+t}{t} > 0$.

For every other numerator with upper argument $a \ge 0$ and requested digit level $k \le x$, $\binom{a}{j} \le \max(1,a)^x \le H^x$. Since $q = B^x H^x$ and $B \ge 2$, the same strict remainder bound follows. There is no derived-label numerator needing a separate degree estimate here. The ratios prove the necessary denominator divisibilities, and the decoded digit equations yield the nine retained range predicates.

7 Soundness and completeness of the new front

7.1 From an integer zero to a finite refutation

The previous sections recover every sign, power, binomial digit, and local range from an arbitrary integer zero. The selector theorem and (9) supply exactly t valid local proof steps. Their beta states join because the same x, y determine the unique remainders modulo $1 + (i+1)y$.

The possible boundary $y = F$ must not be overlooked. Then the first beta remainder is zero, whereas the last is one. Every valid local weight equation propagates a zero state to zero, since all weights are positive. This contradiction proves $y > F$. Thus the initial and terminal states are respectively $F + 1 = w(1)$ and 1.

Multiplying the local equations and cancelling the nonzero intermediate states gives equality between the multiset of conclusion weights and the initial falsity weight together with all premise weights. There are at most t nonunit conclusion factors and at most $2t+1$ nonunit premise factors. All local labels are at most L by Lemma 3.1. Lemma 3.2 gives equality of the corresponding label multisets.

Finally, a balanced single-conclusion rule flow implies a finite derivation of its distinguished target. To see this without assuming the flow is acyclic, close the finitely many displayed rules under derivability, starting with the valid axiom and truth conclusions. If falsity were outside that closure, every rule with an outside conclusion would have at least one outside premise. Summing "outside conclusions minus outside premises" over all occurrences would be at most zero. The recovered balance makes the sum exactly one, contributed by the target falsity. Contradiction. The finite closure computation itself extracts a derivation.

7.2 From a finite refutation to an integer zero

Start with a finite backward proof-tree expansion. Choose the cofinal horizon $t = 128 + 4z^2$ larger than its length and every local schema parameter, including all freshness witnesses. There is no need to make

t exceed the full expanded formula codes. Set $K = t^4$ and $F = t^K$; form all finite pending-multiset products with weights $1 + Fj$, and pad the transcript with truth steps to length t.

Let O be the odd part of $t!$. Choose an arbitrarily large positive odd multiple w of O and put $y = Fw$. Since $v_2(F) = Kv_2(t) \geq 2t > v_2(t!)$, one has $t! \mid y$. Choose w large enough that $y > F$ and every pending state is at most y. The moduli $1 + (i+1)y$, $0 \leq i \leq t$, are pairwise coprime: a common prime would divide both y and one of these moduli after taking an index difference at most t. CRT gives x with the prescribed state remainders; its residue class is cofinal, so all needed endpoint and quotient inequalities can be met.

Now set $B = \binom{F+t}{t}$ and $N = B^x$. At every full prime power $p^e \| B$, select one compatible row and choose all nine representatives to match that row modulo p^{2ex}. The generalized CRT theorem justifies this construction at every prime, including primes not used as the distinguished selector of any row. Polynomial evaluation preserves the local congruences. For a range with local value $w < k$, the falling factorial contains the factor $A - w$, divisible by p^{2ex}; the denominator loses at most $v_p(k!) \leq k - 1 < x \leq ex$ powers. Thus the required binomial coefficient is divisible by p^{ex}.

Choose the eight ordinary representatives first. Then choose C even, positive, and larger than every other nominal index, in its unchanged class modulo the odd integer N^2. Define H and q from their actual values, so enlargement of C strengthens the radix bounds. The ten true normalized remainders are each less than $M/32$. Their common slack in (12) is nonnegative, because $4 \cdot 10/32^2 < 1$. Every integer $n \geq 0$ has the form $r^2 + s^2 + w(w+1)$: apply the three-square theorem to $4n + 1$, whose representation has one odd and two even coordinates.

Choose the large positive Pell center from the square frame, all canonical pairs, and the even-C auxiliary witnesses of Proposition 5.1. The canonical congruences fill the individual and pooled power quotients. This constructs an integer zero of the complete 221-variable quadratic source. The exact projections below carry it to the final equation.

8 Exact quadratic projections and coefficient control

The semantic source has 221 coordinates and 160 quadratic residuals. Its polynomial already has degree four and 7,361 collected monomials, with height 128,446. All remaining reductions are exact algebra over $\mathbb{Z}$, with explicit maps in both directions.

Write a quadratic residual as $r_i(x) = q_i(x) + \ell_i(x) + c_i$. If an integral direction v lies in the common nullspace of all Hessians of the q_i, then

$$r_i(x + sv) = r_i(x) + s\ell_i(v).$$

A coordinate of v equal to ± 1 permits an integral unimodular change of coordinates using v as a column. If some $\ell_i(v) = \pm 1$, that residual solves the new coordinate as a quadratic polynomial in the others. Substitution into the remaining rows preserves degree at most two because the new coordinate occurs linearly with a constant coefficient, namely $\ell_i(v)$, in each remaining row. The inverse map is integral. This gives an existential equivalence over all integers, not merely a rational projection.

Five such projections reduce the source to 216 variables and 155 rows. One more affine consequence gives 215 variables and 154 rows. In the 216-stage coordinates, its exact relation is

$$R_{18} - R_5 = 4 + 12x_8 + 16x_9 + 4x_{19} - x_{20} + x_{33}.$$

Solving for x_{33} and removing the implied row is integral and degree preserving. This immediate chart has large coefficients, but that intermediate cost is removed by the reversible reductions described

next. All these operations are replays of literal certificates; the inherited projection checker reconstructs forward and inverse identities independently of the discovery algorithm.

Finally, integral Pell coordinate changes, integer row additions, and integer translations reduce support and coefficient height. If U is a unimodular row matrix, the common zero set of R equals that of UR; the sums of their squares need not be the same polynomial. The resulting principal equation has 215 variables, 154 residuals, 5,660 collected monomials, and coefficient height 91,679. Its residual coefficient height is 192. Thus the extra affine projection does not require accepting the large coefficient height of its first chart. All reported final monomial counts and heights belong to the fully collected polynomial after these changes.

9 Companions and an explored alternative

The package also provides 213- and 214-variable quartic companions, obtained by choosing $K = 2^{20}t$ and $t = 16 + 4z^2$. Lemma 3.2 still applies, including its boundary. Substituting K directly into the square frame gives the 213-variable version with large coefficients. Retaining K as one separate affine coordinate gives the 214-variable version and reduces that cost. The principal 215-variable version instead uses two quadratic wires for $K = t^4$ and attains the lower coefficient height reported above. The comparison table gives these alternatives under the same unrestricted integer domain and total-degree conventions.

A separate logical experiment splits the old guarded seed bundle into guarded membership vacuity and the one-parameter schema

$$N(\mathrm{nks}(0, 1, c + 2)).$$

The new axioms are old consequences. Conversely, the inherited full-seed completeness proof uses the seed only at roles $0, 1$ with a private code greater than one, while membership vacuity supplies the remaining logical interface. The accompanying proof note explains this ordinary calibration. Under square-shell recoding, the parameter $c + 2$ must be transported by recoding the private code and subtracting two; addition itself is not preserved by the code bijection.

This experiment lowers the formula-depth budget but its independent compiler produced 229 variables where the matched old source uses 225. It loses some sharing among formula constructors. It is retained as a proved research observation, not presented as a smaller principal equation. The principal result uses the original calibrated C_{222} calculus.

10 Logical calibration and what was verified

The semantic chain is

$$\begin{aligned} &\text{integer zero of the printed polynomial} \\ \iff &\text{integer zero of the new arithmetic source} \\ \iff &\text{finite square-shell } C_{222} \text{ refutation} \\ \iff &\text{finite original } C_{222} \text{ refutation} \\ \iff &\text{finite ZF refutation} \iff \text{finite ZFC refutation.} \end{aligned}$$

The last two equivalences are the named repository interfaces `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The latter removes Choice using the constructible-inner-model argument and soundness/completeness. The former uses the seed-model

lower chain and the direct upper interpretation [8]. These endpoint source files were inspected, but their entire transitive Lean dependency closure was not rebuilt in this iteration.

The new contribution is an ordinary mathematical proof, formalizable in ZFC, combined with exact finite computations. The archive's verification driver checks the new source against independently reconstructed semantic formulas, all sixteen power requests and their frame atoms, the signed Pell offsets, the relocated anchor, the removed normalized remainders, every exact projection, and all final coordinate and row changes. It also collects every quartic coefficient and checks the printed appendix against the machine-readable residuals. A rebuild runs in a fresh temporary directory. The same substantive audits run with Python optimization enabled, so correctness checks do not depend on removable `assert` statements.

The inherited unbounded Pell classification, squared-divisibility and comparison-index lemmas, the local-polynomial degree ledger, and the C_{222}/ZF/ZFC calibration remain mathematical dependencies. Their locations and copied proof sources are identified in the provenance files. Finite checking is not described as proving these dependencies anew.

11 Further research questions

1. Can the existing exponent set provide a sufficiently large weight parameter without either the extra K index or the two t^4 wires? A useful candidate must preserve the acyclic order: F precedes the beta code, which precedes the representative C and radix.
2. Can the principal 215-variable form admit another integral projection while keeping its coefficients small? Its common Hessian radical is zero and its 154 quadratic row parts are linearly independent. Both properties persist under invertible affine coordinate changes and constant invertible row operations, so these two projection methods are exhausted. Other polynomial substitutions and different arithmetic encodings remain possible.
3. Can a logic-specific encoding realize the smaller fixed-role seed depth while recovering the arithmetic sharing lost by its two-schema split? The present negative measurement gives a concrete comparator.
4. Can the anchor-transport theorem be formulated once for a general family of arithmetic compilers, with a machine-checked side condition certifying an unbounded CRT representative class and an available canonical pair?
5. The best next formalization target is the new source-to-transcript equivalence and its exact certificate replay. This is more decisive than additional bounded tests of Pell identities. A future complete endpoint should compose it with a reproducible build of the inherited ZFC calibration.

A The literal principal equation

All variables $x_0, \dots, x_{214}$ range independently over $\mathbb{Z}$. The equation is $\sum_{j=0}^{153} R_j^2 = 0$ with the following complete residual list. A missing monomial has coefficient zero. The displayed expressions are parsed back to integer polynomials by the release checker. Coordinate names in construction-stage metadata are historical; the exact inverse certificates give their semantic meaning.

$$R_0 = x_0 - x_0x_4 - x_4 + x_7$$

$$R_1 = -x_2 - x_4x_5 + x_6x_7$$

$$R_2 = -1 + x_1 - x_6x_7$$

$$R_3 = -2x_2x_{213} - x_3x_{18} + x_4 + 2x_{16}$$

$$R_4 = -2 - x_8 - x_8^2 - 2x_8x_9 - 2x_9 - x_9^2 + x_{19}$$

$$R_5 = -2 - x_8 - x_8^2 - 8x_8x_{19} - 4x_8x_{20} - 8x_{19} - 16x_{19}^2 - 16x_{19}x_{20} - 4x_{20} - 4x_{20}^2 + x_{21}$$

$$R_6 = 1 + 2x_{21} - 4x_{21}^2 + x_{22}$$

$$R_7 = -2 - x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{23}$$

$$R_8 = -1 + 2x_8 - x_8^2 - 2x_8x_{10} + 2x_{10} - x_{10}^2 + x_{24}$$

$$R_9 = -7 + 33x_8 - 35x_8^2 - 96x_8x_{10} - 24x_8x_{19} - 4x_8x_{23} - 24x_8x_{24} + 16x_9 - 2x_9^2 - 8x_9x_{23} + 32x_{10} - 64x_{10}^2 - 32x_{10}x_{19} - 32x_{10}x_{24} + 4x_{19} - 4x_{19}^2 - 8x_{19}x_{24} + 12x_{23} - 4x_{23}^2 + 8x_{24} - 4x_{24}^2 + x_{25}$$

$$R_{10} = -16 + 14x_8 - 2x_8^2 - 8x_8x_9 - 8x_8x_{23} + 8x_9 + 8x_{23} - 4x_{25}^2 + x_{26} - 2x_{43}$$

$$R_{11} = -6 + 16x_{10} - 16x_{10}^2 - 16x_{10}x_{23} - 16x_{10}x_{24} + 8x_{23} - 4x_{23}^2 - 8x_{23}x_{24} + 8x_{24} - 4x_{24}^2 + x_{27}$$

$$R_{12} = -5 + 24x_{10} - 16x_{10}^2 - 16x_{10}x_{19} - 32x_{10}x_{24} - 16x_{10}x_{27} + 10x_{19} - 4x_{19}^2 - 16x_{19}x_{24} - 8x_{19}x_{27} + 24x_{24} - 16x_{24}^2 - 16x_{24}x_{27} + 12x_{27} - 4x_{27}^2 + x_{28}$$

$$R_{13} = -12 - 29x_8 - 37x_8^2 - 24x_8x_9 - 112x_8x_{10} - 4x_8x_{23} - 24x_8x_{24} - 39x_9 - 37x_9^2 - 112x_9x_{10} - 24x_9x_{23} - 4x_9x_{24} - 80x_{10} - 128x_{10}^2 - 32x_{10}x_{23} - 32x_{10}x_{24} - 12x_{23} - 4x_{23}^2 - 8x_{24} - 4x_{24}^2 + x_{29}$$

$$R_{14} = -18 - 10x_8 - 2x_8^2 - 24x_8x_9 - 32x_8x_{10} - 8x_8x_{23} - 72x_9 - 72x_9^2 - 192x_9x_{10} - 48x_9x_{23} - 96x_{10} - 128x_{10}^2 - 64x_{10}x_{23} - 24x_{23} - 8x_{23}^2 - 4x_{29}^2 + x_{30}$$

$$R_{15} = -32 - 42x_8 + 8x_8x_{10} + 16x_8x_{23} - 84x_9 + 16x_9x_{10} + 32x_9x_{23} + 21x_{10} - 3x_{10}^2 - 4x_{10}x_{19} - 8x_{10}x_{23} + 21x_{19} - 8x_{19}x_{23} + 24x_{23} - 4x_{23}^2 + x_{31} - x_{59}$$

$$\begin{aligned}R_{16} = &-1-24x_8-64x_8^2-160x_8x_{10}-64x_8x_{23}-32x_8x_{24}\\&-32x_8x_{31}-26x_{10}-100x_{10}^2-80x_{10}x_{23}-40x_{10}x_{24}-40x_{10}x_{31}\\&-8x_{23}-16x_{23}^2-16x_{23}x_{24}-16x_{23}x_{31}-6x_{24}-4x_{24}^2\\&-8x_{24}x_{31}-4x_{31}-4x_{31}^2+x_{32}\end{aligned}$$

$$R_{17} = 1-4x_{19}^2+x_{20}$$

$$\begin{aligned}R_{18} = &-2-x_8-x_8^2+8x_8x_{19}-4x_8x_{20}+8x_{19}\\&-16x_{19}^2+16x_{19}x_{20}-4x_{20}-4x_{20}^2+x_{33}\end{aligned}$$

$$R_{19} = 1+2x_{33}-4x_{33}^2+x_{34}$$

$$R_{20} = -x_9^2-4x_9x_{34}-4x_{34}^2+x_{35}$$

$$\begin{aligned}R_{21} = &-8-20x_8-16x_8^2-64x_8x_9+32x_8x_{19}-40x_9\\&-64x_9^2+64x_9x_{19}+18x_{19}-16x_{19}^2+x_{36}\end{aligned}$$

$$\begin{aligned}R_{22} = &\,8x_9-62x_9^2-92x_9x_{10}-32x_9x_{23}-32x_9x_{36}+6x_{10}\\&-34x_{10}^2-24x_{10}x_{23}-24x_{10}x_{36}-4x_{23}^2-8x_{23}x_{36}+4x_{36}\\&-4x_{36}^2+x_{37}\end{aligned}$$

$$\begin{aligned}R_{23} = &\,8x_8-64x_8^2-96x_8x_{10}-32x_8x_{24}-32x_8x_{37}+6x_{10}\\&-36x_{10}^2-24x_{10}x_{24}-24x_{10}x_{37}+2x_{24}-4x_{24}^2-8x_{24}x_{37}\\&+4x_{37}-4x_{37}^2+x_{38}\end{aligned}$$

$$R_{24} = -x_{10}^2-2x_{10}x_{11}-x_{11}^2+x_{39}$$

$$\begin{aligned}R_{25} = &-32+22x_8-3x_8^2-8x_8x_9-8x_8x_{23}-8x_8x_{24}\\&-4x_8x_{39}+16x_9-8x_9x_{24}-2x_{10}^2-8x_{10}x_{39}+16x_{23}\\&-8x_{23}x_{24}+22x_{24}-4x_{24}^2+4x_{39}-4x_{39}^2+x_{40}\\&-x_{43}\end{aligned}$$

$$\begin{aligned}R_{26} = &\,6-16x_8^2-36x_8x_{10}-32x_8x_{24}-40x_8x_{39}-16x_8x_{40}\\&-22x_{10}^2-32x_{10}x_{24}-48x_{10}x_{39}-16x_{10}x_{40}-2x_{24}-16x_{24}^2\\&-32x_{24}x_{39}-16x_{24}x_{40}+4x_{39}-24x_{39}^2-16x_{39}x_{40}-4x_{40}^2\\&+x_{41}\end{aligned}$$

$$\begin{aligned}R_{27} = &\,x_{10}-25x_{10}^2-40x_{10}x_{24}-20x_{10}x_{27}-16x_{24}^2-16x_{24}x_{27}\\&-4x_{27}^2+x_{42}\end{aligned}$$

$$\begin{aligned}R_{28} = &-1+8x_9-4x_9^2-8x_9x_{23}+8x_{23}-4x_{23}^2\\&+x_{43}\end{aligned}$$

$$R_{29} = -1 - 16x_9^2 - 8x_9x_{10} - 32x_9x_{23} - 16x_9x_{43} - x_{10}^2 - 8x_{10}x_{23} - 4x_{10}x_{43} - 16x_{23}^2 - 16x_{23}x_{43} - 4x_{43}^2 + x_{44}$$

$$R_{30} = 3 + x_9 - 2x_9x_{10} - 4x_9x_{44} - 3x_{10}^2 - 8x_{10}x_{44} - x_{23} - 4x_{44}^2 + x_{45}$$

$$R_{31} = -1 - 4x_{45} - 4x_{45}^2 + x_{46}$$

$$R_{32} = -1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} - 16x_{45}^2 - 16x_{45}x_{46} - 4x_{46}^2 + x_{47}$$

$$R_{33} = 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{34} = -4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49}$$

$$R_{35} = -9 + 7x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} + 24x_{42} - 16x_{42}^2 - 16x_{42}x_{49} + 12x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = -x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{52} - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -10 - 12x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = -19 + 4x_9 - 4x_9^2 - 8x_9x_{23} - 16x_9x_{53} - 8x_9x_{54} + 4x_{23} - 4x_{23}^2 - 16x_{23}x_{53} - 8x_{23}x_{54} - 16x_{53} - 24x_{53}^2 - 16x_{53}x_{54} + 6x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{41} = -10 + x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{55} - 16x_{23}^2 - 16x_{23}x_{55} - 12x_{48} - 4x_{48}^2 - 4x_{55}^2 + x_{56}$$

$$R_{42} = 21 + 24x_{48} - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57}$$

$$R_{43} = -5 + 4x_8 + 8x_9 + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 2x_{19} - 4x_{57}^2 + x_{58} + 12x_{59} - 4x_{59}^2$$

$$\begin{aligned} R_{44} = & -6x_8 - 16x_8^2 - 64x_8x_9 + 8x_8x_{10} + 16x_8x_{19} - 12x_9 \\ & - 64x_9^2 + 16x_9x_{10} + 32x_9x_{19} + 3x_{10} - x_{10}^2 - 4x_{10}x_{19} \\ & + 3x_{19} - 4x_{19}^2 + x_{59} \end{aligned}$$

$$\begin{aligned} R_{45} = & -12 + 8x_8 + 16x_9 - 4x_{19} - 4x_{58}^2 + 24x_{59} \\ & - 8x_{59}^2 + x_{60} \end{aligned}$$

$$R_{46} = -1 - 4x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{47} = -x_2x_{13} + x_{62}$$

$$R_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$R_{49} = x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$R_{50} = x_{12} - x_{13} - 2x_{28}x_{62} - 6x_{62} - x_{65}$$

$$R_{51} = x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$R_{52} = x_{12} - x_{13} - 2x_{32}x_{62} + 2x_{62} - x_{67}$$

$$R_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$R_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$R_{55} = 2x_{28}x_{62} - 2x_{41}x_{62} - 6x_{62} + x_{65} - x_{70}$$

$$\begin{aligned} R_{56} = & -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 \\ & + x_{71} \end{aligned}$$

$$R_{57} = 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72}$$

$$\begin{aligned} R_{58} = & -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 \\ & + x_{73} - x_{74}^2 \end{aligned}$$

$$R_{59} = -4x_{60}x_{62} - 2x_{61}x_{62} + x_{74}$$

$$R_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$

$$R_{61} = -1 + 4x_2 + 4x_2x_8 + 8x_2x_9 - 2x_2x_{19} + x_{75}$$

$$R_{62} = -x_8x_{208} - x_{12} + x_{76} + x_{208}$$

$$R_{63} = x_9x_{208} - 2x_{19}x_{62} + 2x_{32}x_{62} + x_{67} + x_{78}$$

$$R_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$

$$R_{65} = -x_{64}x_{79} + x_{80}$$

$$R_{66} = -x_{65}x_{80} + x_{81}$$

$$R_{67} = -x_{66}x_{81} + x_{82}$$

$$R_{68} = -x_{67}x_{82} + x_{83}$$

$$R_{69} = -x_{68}x_{83} + x_{84}$$

$$R_{70} = -x_{69}x_{84} + x_{85}$$

$$R_{71} = -x_{70}x_{85} + x_{86}$$

$$R_{72} = -x_{77}x_{86} + x_{87}$$

$$R_{73} = -x_{78}x_{87} + x_{88}$$

$$R_{74} = -x_{88}^2 + x_{89}$$

$$R_{75} = -x_4 - 2x_4x_{16} + 2x_{16} + x_{90}$$

$$R_{76} = -x_{14}x_{90} + x_{91}$$

$$\begin{aligned} R_{77} = {} & -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 \\ & - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{94} - x_{73}x_{89} - x_{91}^2 \\ & - x_{92}^2 \end{aligned}$$

$$R_{78} = -x_4x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$R_{79} = -x_{95}x_{98} + x_{98} + x_{100}$$

$$R_{80} = -x_3x_{121} + x_{96}x_{200} - x_{99} - x_{100}x_{121}$$

$$R_{81} = x_{101}x_{200} - x_{102}x_{124} - x_{103}$$

$$R_{82} = x_{104}x_{200} - x_{105}x_{124} - x_{106}$$

$$R_{83} = x_{107}x_{200} - x_{108}x_{124} - x_{109}$$

$$R_{84} = x_{110}x_{200} - x_{111}x_{124} - x_{112}$$

$$R_{85} = x_{113}x_{201} - x_{115}x_{124} - x_{116}$$

$$R_{86} = x_{117}x_{201} - x_{118}x_{124} - x_{119}$$

$$R_{87} = -x_{17}x_{121} + x_{124}$$

$$R_{88} = x_{120} - x_{122}x_{124} - x_{123}$$

$$R_{89} = -x_{124}x_{126} + x_{125} - x_{127}$$

$$R_{90} = -x_{124}x_{129} + x_{128}x_{200} - x_{130}$$

$$R_{91} = x_{131} - 2x_{131}x_{133} + x_{134}$$

$$R_{92} = -x_{131}x_{134} - x_{134} + x_{135}$$

$$R_{93} = -1 + x_{132}^2 + x_{134}^2 - x_{135}^2$$

$$R_{94} = 1 - x_{132}x_{137} - x_{136} + x_{138}$$

$$R_{95} = -1 + 2x_{136} - x_{136}^2 + 2x_{136}x_{138} + x_{137}^2$$

$$R_{96} = 1 + x_0^2 - 2x_0x_{132} + x_{139}$$

$$R_{97} = -x_0x_{182} + x_2 + x_{139}x_{140} - x_{181}$$

$$R_{98} = 1 + x_3^2 - 2x_3x_{132} + x_{141}$$

$$R_{99} = -x_3x_{173} + x_{17} + x_{141}x_{142} - x_{172}$$

$$R_{100} = -1 + 2x_{132}x_{184} - x_{143} - x_{184}^2$$

$$R_{101} = -1 + x_{95} + x_{143}x_{144} - x_{172} - x_{173}x_{184}$$

$$R_{102} = -x_{132}x_{146} - x_{145} + x_{147}$$

$$R_{103} = -1 - x_{145}^2 + 2x_{145}x_{147} + x_{146}^2$$

$$R_{104} = 1 - x_{132}x_{149} - x_{148} + x_{150}$$

$$R_{105} = -1 + 2x_{148} - x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$R_{106} = -x_{132}x_{152} - x_{151} + x_{153}$$

$$R_{107} = -1 - x_{151}^2 + 2x_{151}x_{153} + x_{152}^2$$

$$R_{108} = -x_{132}x_{155} - x_{154} + x_{156}$$

$$R_{109} = -1 - x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$R_{110} = -x_{132}x_{158} - x_{157} + x_{159}$$

$$R_{111} = -1 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$R_{112} = -x_{132}x_{161} - x_{160} + x_{162}$$

$$R_{113} = -1 - x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$

$$R_{114} = 1 - x_{132}x_{164} - x_{163} + x_{165}$$

$$R_{115} = -1 + 2x_{163} - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$R_{116} = -x_{132}x_{167} - x_{166} + x_{168}$$

$$R_{117} = -1 - x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$R_{118} = -x_{132}x_{170} - x_{169} + x_{171}$$

$$R_{119} = -1 - x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$

$$R_{120} = 1 - x_{132}x_{173} - x_{172} + x_{174}$$

$$R_{121} = -1 + 2x_{172} - x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$R_{122} = -x_{132}x_{176} - x_{175} + x_{177}$$

$$R_{123} = -1 - x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$R_{124} = -x_{132}x_{179} - x_{178} + x_{180}$$

$$R_{125} = -1 - x_{178}^2 + 2x_{178}x_{180} + x_{179}^2$$

$$R_{126} = 1 - x_{132}x_{182} - x_{181} + x_{183}$$

$$R_{127} = -1 + 2x_{181} - x_{181}^2 + 2x_{181}x_{183} + x_{182}^2$$

$$\begin{aligned} R_{128} = & -5 - 2x_0 - 4x_0^2 - 2x_0x_2 - x_1^2 - 2x_2^2 \\ & - x_3^2 - x_4^2 + 2x_8 - x_8^2 - x_9^2 - x_{10}^2 \\ & - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - 4x_{16}^2 \\ & - x_{17}^2 - x_{93}^2 + 2x_{95} - 2x_{95}^2 - x_{96}^2 - x_{97}^2 \\ & - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 - x_{114}^2 \\ & - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 + x_{131} \\ & - x_{184}^2 \end{aligned}$$

$$\begin{aligned} R_{129} = & 2x_3 - 4x_3x_8 - 4x_3x_9 - 4x_3x_{10} - 4x_3x_{11} - 4x_3x_{12} \\ & - 4x_3x_{13} - 4x_3x_{14} - 4x_3x_{15} - 8x_3x_{16} + x_{184} \end{aligned}$$

$$R_{130} = -128 + x_0 - 4x_{185}^2$$

$$R_{131} = -x_{132}x_{187} + x_{196}$$

$$R_{132} = -1 + x_{186}^2 + x_{187}^2 - x_{196}^2$$

$$R_{133} = -x_{189}x_{190} + x_{197}$$

$$R_{134} = -1 + x_{188}^2 + x_{189}^2 - x_{197}^2$$

$$R_{135} = -x_{179}^2 + x_{198}$$

$$R_{136} = x_{187} - 2x_{192}x_{198} - x_{198}$$

$$R_{137} = -1 - 4x_{187} - 8x_{187}x_{191} + x_{190}$$

$$R_{138} = -x_{132} - x_{186}x_{193} + x_{190}$$

$$R_{139} = -x_{180} - x_{186}x_{194} + x_{188}$$

$$R_{140} = -2x_{16} - 4x_{179}x_{195} + x_{189}$$

$$\begin{aligned} R_{141} = & 9 + 2x_0^2 + 2x_0x_2 - 2x_0x_{137} - 2x_0x_{146} + x_1^2 \\ & - 2x_1x_{173} + x_2^2 - 2x_2x_{146} + x_4^2 - 2x_4x_{164} - 2x_8 \\ & + x_8^2 - 2x_8x_{149} + x_9^2 - 2x_9x_{152} + x_{10}^2 - 2x_{10}x_{155} \\ & + x_{11}^2 - 2x_{11}x_{158} + x_{12}^2 - 2x_{12}x_{161} + x_{13}^2 - 2x_{13}x_{167} \\ & + x_{14}^2 - 2x_{14}x_{170} + x_{15}^2 - 2x_{15}x_{176} - 4x_{16}x_{179} + x_{93}^2 \\ & - 2x_{93}x_{182} - x_{132}x_{199} + x_{138}^2 - 2x_{147} + x_{147}^2 + 2x_{149} \\ & + x_{150}^2 - 2x_{153} + x_{153}^2 - 2x_{156} + x_{156}^2 - 2x_{159} \\ & + x_{159}^2 - 2x_{162} + x_{162}^2 + x_{165}^2 - 2x_{168} + x_{168}^2 \\ & - 2x_{171} + x_{171}^2 + x_{174}^2 - 2x_{177} + x_{177}^2 + 2x_{179}^2 \\ & - 2x_{180} + x_{180}^2 + x_{183}^2 + x_{199} - x_{202}^2 - x_{203}^2 \\ & - x_{204} - x_{204}^2 \end{aligned}$$

$$R_{142} = -x_{97}x_{200} + x_{121}$$

$$R_{143} = -x_{114}x_{201} + x_{121}$$

$$\begin{aligned} R_{144} = & -1 - 4x_{16}^2 - x_{137}^2 - x_{146}^2 - x_{149}^2 - x_{152}^2 \\ & - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 - x_{170}^2 \\ & - x_{173}^2 - x_{176}^2 + x_{179}^2 - x_{182}^2 - x_{202}^2 - x_{203}^2 \\ & - x_{204} - x_{204}^2 \end{aligned}$$

$$\begin{aligned} R_{145} = & -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 \\ & - 4x_{116}^2 - 4x_{119}^2 + x_{121}^2 - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 \\ & - x_{205}^2 - x_{206}^2 - x_{207} - x_{207}^2 \end{aligned}$$

$$R_{146} = -x_2x_{12} + x_{208}$$

$$\begin{aligned} R_{147} = & x_{95}x_{210} - 2x_{96}x_{146} - 2x_{101}x_{149} - 2x_{104}x_{152} - 2x_{107}x_{155} - 2x_{110}x_{158} \\ & - 2x_{113}x_{161} - 2x_{117}x_{167} - 2x_{120}x_{170} - 2x_{125}x_{176} - 2x_{128}x_{179} - 2x_{132}x_{210} \\ & + 2x_{146}x_{147} + 2x_{149} + 2x_{149}x_{150} + 2x_{152}x_{153} + 2x_{155}x_{156} + 2x_{158}x_{159} \\ & + 2x_{161}x_{162} + 2x_{167}x_{168} + 2x_{170}x_{171} + 2x_{176}x_{177} + 2x_{179}x_{180} + x_{209} \end{aligned}$$

$$\begin{aligned} R_{148} = & -10 - x_{95}x_{209} + x_{96}^2 - 2x_{96}x_{145} + x_{101}^2 - 2x_{101}x_{148} \\ & + x_{104}^2 - 2x_{104}x_{151} + x_{107}^2 - 2x_{107}x_{154} + x_{110}^2 - 2x_{110}x_{157} \\ & + x_{113}^2 - 2x_{113}x_{160} + x_{117}^2 - 2x_{117}x_{166} + x_{120}^2 - 2x_{120}x_{169} \\ & + x_{125}^2 - 2x_{125}x_{175} + x_{128}^2 - 2x_{128}x_{178} + 2x_{145}x_{147} + 2x_{148} \\ & + 2x_{148}x_{150} + 2x_{151}x_{153} + 2x_{154}x_{156} + 2x_{157}x_{159} + 2x_{160}x_{162} + 2x_{166}x_{168} \\ & + 2x_{169}x_{171} + 2x_{175}x_{177} + 2x_{178}x_{180} + x_{210} \end{aligned}$$

$$\begin{aligned} R_{149} = & x_{95}x_{212} - 2x_{97}x_{137} - 2x_{114}x_{164} - 2x_{121}x_{173} - 2x_{132}x_{212} + 2x_{137} \\ & + 2x_{137}x_{138} + 2x_{164} + 2x_{164}x_{165} + 2x_{173} + 2x_{173}x_{174} + x_{211} \\ & - x_{212} \end{aligned}$$

$$\begin{aligned} R_{150} = & -2 - x_{95}x_{211} + x_{97}^2 - 2x_{97}x_{136} + x_{114}^2 - 2x_{114}x_{163} \\ & + x_{121}^2 - 2x_{121}x_{172} + 2x_{136} + 2x_{136}x_{138} + 2x_{163} + 2x_{163}x_{165} \\ & + x_{172}^2 - x_{173}^2 + x_{211} + x_{212} \end{aligned}$$

$$R_{151} = -1 - x_2 - 2x_2x_{213} + x_4$$

$$R_{152} = -x_0^2 + x_{214}$$

$$R_{153} = x_{93} - x_{214}^2$$

B Artifact map and reproducibility

Path	Purpose
`report/report.tex`, PDF	Article, ordinary proofs and complete principal residuals.
`equations/principal/`	Final quadratic residuals, collected polynomial, metrics, text and variable list.
`equations/companions/`	Measured alternatives with smaller variable counts and different heights.
`source/parent225/`	Pinned immediate parent and attribution.
`source/anchor221/`	Complete new semantic source and literal change certificate.
`stages/`	Exact projection and coordinate-change boundaries.
`proofs/`	New unbounded proofs and independently assigned AI mathematical audits.
`inherited/`	Identified inherited arithmetic, syntax and logical proof sources.
`code/`, `verify.py`	Deterministic producer, independent replay, evaluation and report checks.
`verification/`	Receipts, exact source identities and finite checks actually run.
`PROVENANCE.json`, `SHA256SUMS`	Pinned origins and release integrity.

Run `python3 -B verify.py` from the extracted package. The verification and polynomial evaluation require only the Python standard library. The PDF can be rebuilt with ordinary pdfLaTeX; timestamps and TeX versions may change PDF bytes without changing any coefficient data. See the README for the exact rebuild commands and recorded verification scope. The provided evaluation program evaluates a specified tuple; it does not decide whether a zero exists.

References

[1] ProveIt, *Pooling exact power certificates for Diophantine consistency*, source 158, 25 October 2026. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/158-pooled-powers`.

[2] ProveIt, *Removing factorial authenticity from Diophantine consistency*, source 151, 25 October 2026. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/151-factorial-free-binomial-crt`.

[3] ProveIt, *Linear CRT radii and semantic radix recovery*, source 157, 25 October 2026. Pinned alongside the preceding source; the copied `MAIN_PROOF.md` supplies the sign bootstrap and every-prime CRT completeness.

[4] ProveIt, *Quadratic projection*, source 159, 25 October 2026. Exact common-Hessian and affine projection algorithms, with an independent integer certificate checker.

[5] ProveIt, *Reusing a semantic Pell anchor over unrestricted integers* and *Recovering nonnegative arithmetic from signed Pell orientation*, source 135, retained in source 158's provenance. Copied as `PROOF_315.md` and `ORIENTATION_FIRST_PROOF.md`.

[6] ProveIt, *Aligning the amplification exponent with a strict beta quotient range*, source 135, and source 158's *Cofinal strict bounds and a linear beta base*.

[7] ProveIt, *Square-shell encoding proof* and the literal `shell_source.py` constructor source; identified by source 158's `SHELL_ENCODING_PROOF.md` supplement and exact source hashes.

[8] ProveIt, Lean sources `KCode222.lean`, `KSeedDirect222.lean`, `KEndpoint.lean`, and `ConstructibleChoice.lean`. Exact origins and inspected copies are listed in `PROVENANCE.json`; no new full Lean build is claimed.